

# **AGILE SOFTWARE ENGINEERING (CS303)**

## **Agile Manifesto Principles**

### **Learning Objectives**

Understand Agile development, explain the Agile Manifesto, describe the four core values, discuss the twelve Agile principles, and identify Agile benefits in modern software engineering.

### **Introduction to Agile**

Agile is an iterative and incremental approach to software development. It focuses on customer collaboration, continuous improvement, adaptability, and delivering working software in small releases. Agile became popular because traditional models struggled with changing requirements.

### **What is the Agile Manifesto?**

The Agile Manifesto was created in 2001 by seventeen software practitioners. It defines a set of values and principles that guide Agile teams in delivering high-quality software.

### **Four Core Values of Agile**

1. Individuals and Interactions over Processes and Tools. 2. Working Software over Comprehensive Documentation. 3. Customer Collaboration over Contract Negotiation. 4. Responding to Change over Following a Plan.

### **Value Explanations**

Agile emphasizes communication, teamwork, customer involvement, and flexibility. While processes, documentation, contracts, and plans remain important, Agile values place greater emphasis on people and adaptability.

### **The Twelve Agile Principles**

Customer satisfaction through early delivery, welcome changing requirements, frequent software releases, collaboration between business and developers, motivated individuals, face-to-face communication, working software as progress measure, sustainable development, technical excellence, simplicity, self-organizing teams, and continuous improvement.

### **Benefits of Agile**

Faster delivery, reduced risk, improved product quality, better stakeholder involvement, greater transparency, and enhanced customer satisfaction.

### **Limitations of Agile**

Requires active customer participation, experienced team members, continuous communication, and may be difficult to scale in very large projects.

## **Agile Frameworks**

Popular Agile frameworks include Scrum, Kanban, Extreme Programming (XP), Lean Development, and Crystal Methods.

## **Real-World Applications**

Companies such as Google, Amazon, Netflix, Spotify, and Microsoft use Agile methodologies to develop products and services.

## **Exam Tips**

Focus on the Agile Manifesto, four values, twelve principles, Agile benefits, Agile limitations, and Agile vs Traditional development models.

## **Practice Questions**

2 Marks: Define Agile. What is Agile Manifesto? 5 Marks: Explain the four core values. 10 Marks: Discuss the twelve Agile principles and their importance.

## **Summary**

Agile is a flexible software development methodology that promotes collaboration, customer involvement, and continuous delivery. The Agile Manifesto provides the foundation for Agile practices used worldwide.